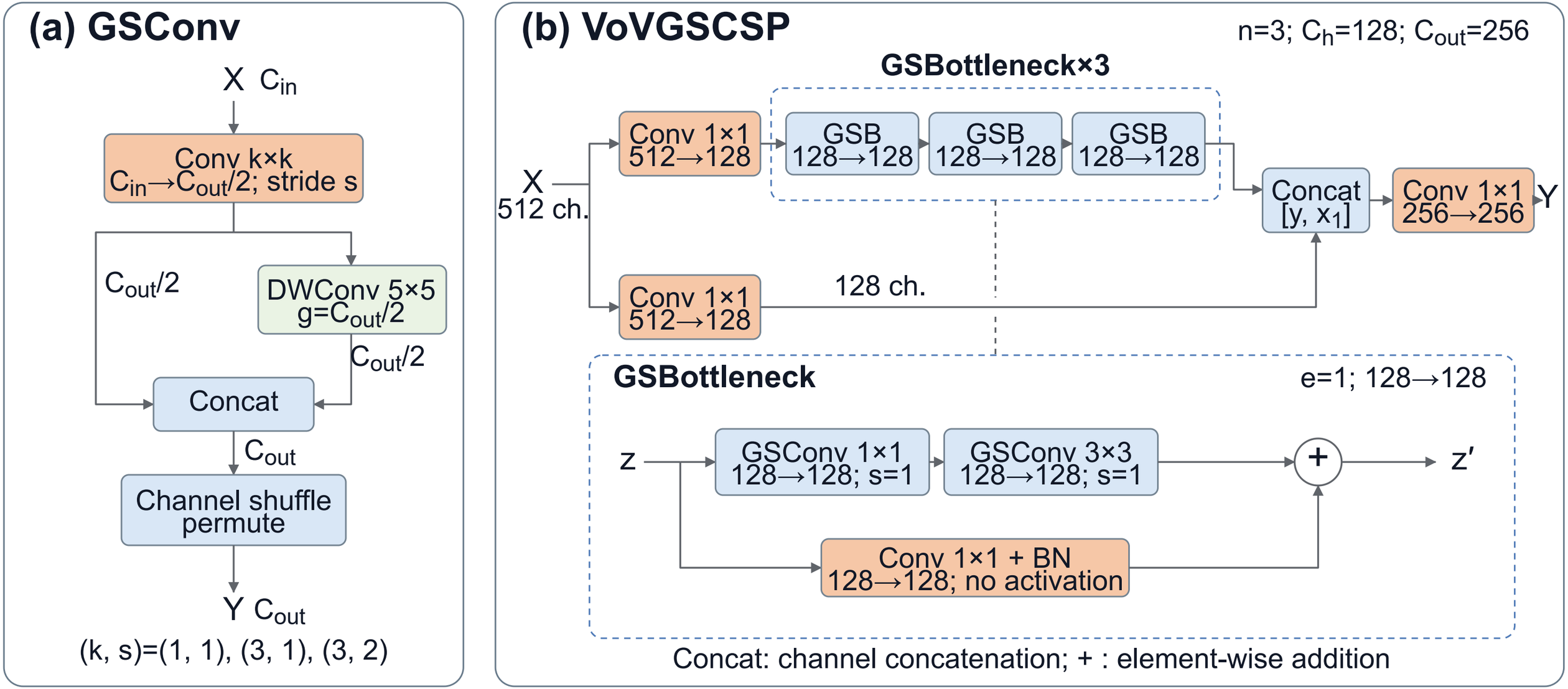

(a) GSConv
X Cin
Conv k×k
Cin→Cout/2; stride s
Cout/2
DWConv 5×5
g=Cout/2
Cout/2
Concat
Cout
Channel shuffle
permute
Y Cout
(k, s)=(1, 1), (3, 1), (3, 2)
(b) VoVGSCSP
n=3; Ch=128; Cout=256
GSBottleneck×3
X
512 ch.
Conv 1×1
512→128
GSB
128→128
GSB
128→128
GSB
128→128
Concat
[y, x1]
Conv 1×1
256→256
Y
Conv 1×1
512→128
128 ch.
GSBottleneck
e=1; 128→128
z
GSConv 1×1
128→128; s=1
GSConv 3×3
128→128; s=1
+
z′
Conv 1×1 + BN
128→128; no activation
Concat: channel concatenation; + : element-wise addition